

TAK Server - Federations

TAK Server - Federations Intro

- Federations allow subsets of TAK users who are connected to different servers to work together, even though each server may be run by independent organizations.
- Each admin domain does not need to share anything about their internal structure with the other admin domains.
- Each admin domain has control over what data they share with the other domains but has no control over what the other admin domains do with the data that is shared.
- It requires no reconfiguration of TAK devices connected to either end and does not allow direct connections of TAK devices from the other admin domains.

TAK Server - Federation Setup

- Enable federation configuration:
 - On the admin dashboard:
 - Navigate to the following: hamburger menu → Configuration → Federation
 - Select “Edit Configuration”
 - Enable “Federation Enabled: “
 - Select “Save”
 - Make note of the port that the federation is using:
 - 9001

Federation Configuration

Federation Enabled: ☒
Federation V1 Enabled: ☐ Core Messaging Version: 2 Port: 9000, TLS Version(s): TLSv1.2, TLSv1.3
Federation V2 Enabled: ☒ Port: 9001 Allow Mission Federation: true Allow DataFeed Federation: true Allow Federated Delete: false Federated Group Mapping: Automatic Group Mapping: Enable Mission Federation Disruption Tolerance: true Mission Federation Disruption Tolerance Interval: 12 hours Clear Federation Events Will resync federated missions on reconnect
Enable Data Package And Mission File Filter: false
Federation Truststore File Path: certs/files/fed-truststore.jks
Web Base URL: https://127.0.0.1:8443/Marti

[Edit Configuration](#)

TAK Server - Federation Setup - Firewall

- Within a terminal that is SSH'd into the server run the following:
 - List all the ports:
 - **sudo firewall-cmd --list-all**
 - If the port we are using for the federation is not listed, add it:
 - **sudo firewall-cmd --zone=public --add-port 9001/tcp --permanent**
 - Reload the firewall
 - **sudo firewall-cmd --reload**
 - Ensure the port update:
 - **sudo firewall-cmd --list-all**

```
[takadmin@localhost tak]$ sudo firewall-cmd --zone=public --add-port 9001/tcp --permanent
success
[takadmin@localhost tak]$ sudo firewall-cmd --reload
success
[takadmin@localhost tak]$ sudo firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3 enp0s8
  sources:
  services: cockpit dhcpv6-client ssh
  ports: 8089/tcp 8443/tcp 8446/tcp 8087/tcp 9001/tcp
  protocols:
  forward: no
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[takadmin@localhost tak]$ _
```

TAK Server - Federation Setup - Upload Federate Certificates

- To connect federate servers, you need to exchange their public CA:
 - CA location:
 - `cd /opt/tak/certs/files/ca.pem`
- Once the CAs have been exchanged you can go to the “Federate Certificate Authorities”:
 - Admin dashboard → hamburger menu → Configuration → Federate Certificate Authorities
 - Select “Upload Federate Certificate Authority”

Federation Certificate Authorities

[Upload Federate Certificate Authority](#)

Issuer	Subject	Serial Number			Max hops
CN=TAK-ROOT-CA-01, OU=TEST, O=TEST, L=XX, ST=XX, C=XX	CN=TAK-ROOT-CA-01, OU=TEST, O=TEST, L=XX, ST=XX, C=XX	6.370028489114345e+47	Delete	Manage CA Groups	-1
CN=TAKSERVER, OU=ANGRY1, O=ANGRY, L=BFE, ST=BFE, C=US	CN=intermediate-ca, OU=ANGRY1, O=ANGRY, L=BFE, ST=BFE, C=US	5374130301205558000	Delete	Manage CA Groups	-1

TAK Server - Federation Setup - Making the Connection

- To make the connection, only one of the servers will create an outgoing connection to the other.
 - On the server that is starting the connection, go to the “Federation” section again.
 - To create the connection:
 - Select “Create Outgoing Connection”
 - Fill in the necessary information:
 - Display Name: <Your choice>
 - Address: <IP address of that server>
 - Port: 9001 (unless using a different one)
 - Select “Save”

Active Connections

Federate Configuration	Remote Address	Port	Initiator	Read Count	Processed Count	Groups	Group Mappings (Remote:Local)	Manage
takserver:intermediate-ca	192.168.1.105	9001	Self	0	0			View Contacts Delete Edit Config Edit Groups Edit Missions

Outgoing Connection Configuration

[Create Outgoing Connection](#)

Name	Address	Port	Fallback Connection	Status	Reconnect Interval	Max Retries	Federate	Protocol Version	Connection Status	Last Error		
Tetsing	192.168.1.105	9001		Enabled (Disable Connection)	30	Unlimited	takserver:intermediate-ca	2	CONNECTED		Edit	Delete

Federate Configuration

Name	Share Alerts	Archive	Notes	
takserver:intermediate-ca	Enabled	Enabled		Delete Edit Config Edit Groups Edit Missions

TAK Server - Federation Setup - Making the Connection

- You will notice that no information is flowing between the federates, this is because both federates need to specify which groups to allow to flow to the other server.
- You can specify the group and the directions of information flowing between the servers.

Federate Group Filtering

You are configuring groups for federate: **testingCL:test-CA**

A federate is another TAK installation with which you wish to share events.

When events arrive from this federate, you may direct them to devices that belong to any of the local inbound groups configured below. Similarly, you may send events to this federate from groups configured.

Note: Adding and Removing groups affects the runtime environment immediately, and changes are saved to the configuration file. This configuration is associated with federate connections connection is initiated.

Group

Search LDAP

Direction

Both (Inbound/Outbound)

Add Group

Groups configured for this federate:

Group	Direction	
__ANON__	INBOUND	Remove
TEST	INBOUND	Remove
__ANON__	OUTBOUND	Remove
TEST	OUTBOUND	Remove

TAK Server - Federation Setup - Making the Connection

- To manage the groups of the federate:
 - Select “Edit Groups” for the desired federate.
 - Federate Group Filtering:
 - If you have LDAP set-up you will be able to search for the group and add it.
 - Federate Group Mapping:
 - You can specify the incoming group and map it to your desired group.
 - Server1(sender): __ANON__ Server2(receiver): __ANON__ → TEST
- You can have multiple group mappings to both multiple remote groups and multiple local groups.

Federated Group Mapping

When group mapping is enabled, the local incoming groups defined above will not be used (unless 'fallback when no group mappings' is enabled). Events that arrive will have the remote groups they originated from attached to them. This allows you to map a specific remote groups directly to any local groups.

The Remote Group List below will only be populated when the federate is connected. Manual entries can be added in the Remote Group field. A remote group may be mapped to many local groups.

Remote Group List

Remote Group

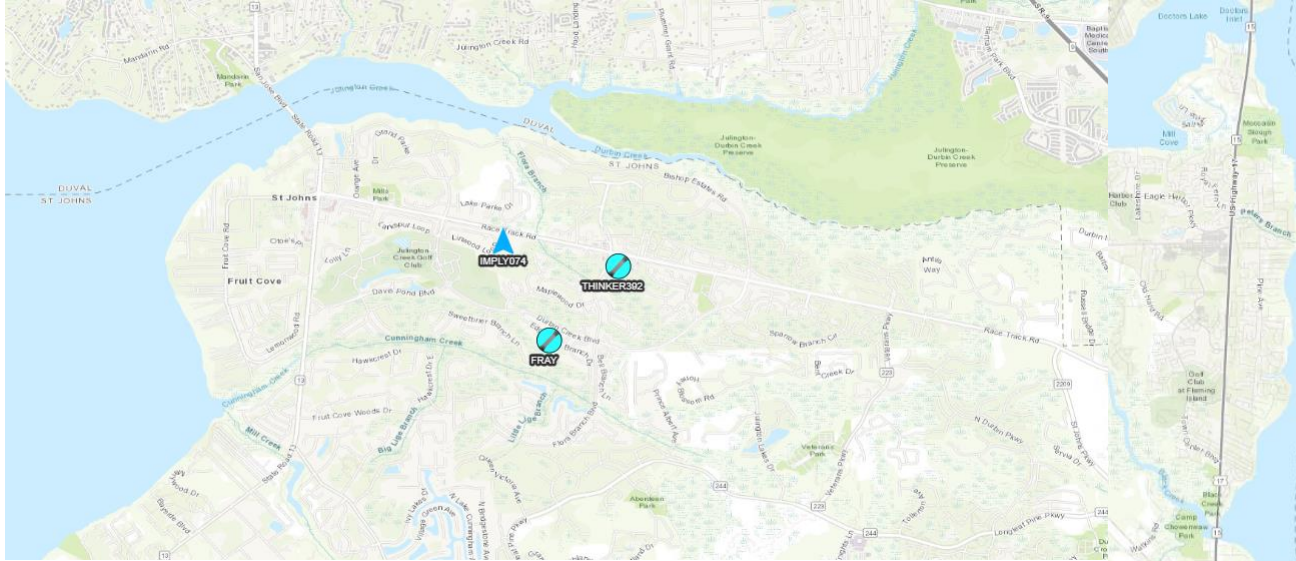
Local Group

Group Mapping configured for this federate:

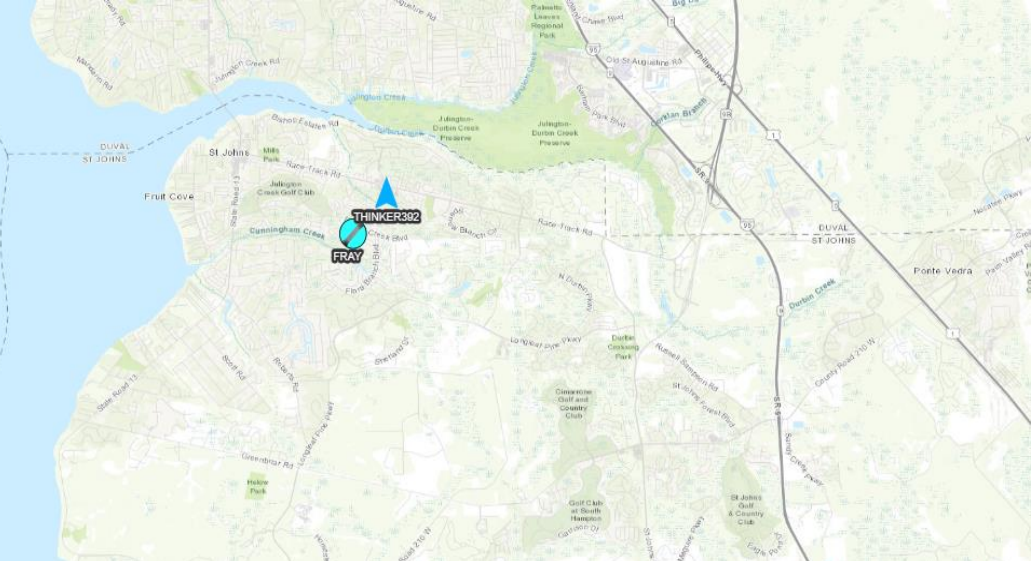
Remote Group	Local Groups	Group Removal
__ANON__	['TEST']	-- Select Group -- Remove

TAK Server - Federation Connection

Server 1: Making Connection



Server 2: Receiving Connection



- Above you can see on the left that the server receiving the federate connection shows the other markers on the map. The right side is showing the other federated marker, along with other markers.

TAK Server - Mission Federation Disruption Tolerance

- Within the “Federation” tab, when editing the federation configuration, there is a “Enable Mission Federation Disruption Tolerance” option.
 - The option will update each server with changes to federated missions that occurred during a disruption in the connection.
 - Below that you are able to limit the changes that are updated to the server to a predetermined length of time.
 - If the “Unlimited” option is selected, then all updates made between the disruption and the time of reconnecting will be loaded.
 - Be aware that without a limit there is a potential for an update to take a lot of bandwidth.
 - There is also a “Clear Federation Events” button that is available when done editing the configuration that will reset the disruption history for federations.

TAK Server - Data Package and Mission File Blocker

- Data packages and mission packages can be federated between servers and their respective clients. These can lead to unwanted configuration changes to TAK devices.
- Within the federation configuration you are able to set a filter to block files by their type.
 - To use this feature:
 - Enable the “Enable Data Packaged and Mission File Filter” option.
 - Enter the file type you wish to block within the “Add File Extension” box then selecting “Add”.
 - By default, the type is: .pref
 - You can change the type to of file that is blocked by adding a new.
 - Once done select “Save” at the bottom.

Data Package and Mission File Filter
Enable Data Package and Mission File Filter: ☒

File Extension: .exe
Add File Extension: The default extension is .pref.
Enter the extension of the file type to block in federated data packages and missions.